
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

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ABSTRACT

Algorithms increasingly function as information mediators, including social-media ranking systems, targeted-advertising pipelines, and LLM-based interfaces. Their deployment induces a coupled decision process in which agents aggregate algorithmically mediated private signals with information inferred from the observed actions of predecessors. We formalize this setting through a controlled sequential social-learning model in which an information-mediating planner (e.g., an LLM) selects the agents’ signal structure while agents continue to update from prior actions. The planner’s objective may be welfare-aligned, in which case it is altruistic, or state-independent and action-targeting, in which case it is biased. The resulting optimization problem combines dynamic programming, decentralized agent choice, and Bayesian public-belief updating. We prove convexity of the altruistic value function and characterize optimal policies for both planner types, thereby specifying the tradeoff between precision-investment costs and rewards induced by agent actions. The characterization partitions the public-belief space into regimes in which the planner invests maximally, does not invest, or chooses precision levels that vary monotonically with belief. In particular, the biased planner may optimally reduce signal precision over some belief ranges, thereby intentionally obfuscating agents’ private information. Even under restrictive transparency constraints—information parity with individuals, no lying or cherry-picking, and full observability—information mediation can substantially increase or decrease social welfare. We complement the theoretical analysis with simulations in which LLMs instantiate both planner and agents. The LLM planner exhibits emergent strategic behavior that broadly matches the predicted policy regimes, while systematic deviations indicate the effect of non-Bayesian reasoning, consistent with cognitive patterns observed in human users and in LLMs trained on human-like data. Overall, the framework provides a tractable technical basis for analyzing and regulating LLM information mediators whose behavior is coupled to social learning.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly embedded as information-mediating components in socially consequential systems, ranging from search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) to personal assistants that provide medical or political guidance (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). Existing evidence indicates that LLMs can behave persuasively (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulatively (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategically (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) during user interactions. At population scale, such algorithmic influence is not isolated from interpersonal information transmission: it interacts with observational learning, through which individuals infer latent information from others’ actions. This coupling can amplify, attenuate, or qualitatively transform the effect of the model-mediated signals.

As a concrete instance, consider an LLM-powered recommendation system that sequentially advises potential customers about a new restaurant. The restaurant’s quality is the binary latent state: it is either good or bad, and this state is unknown to both users and the system. Each user receives a personalized recommendation, interpreted as a private signal, and then chooses whether to patronize

the restaurant. The recommendation system, which we model as the *planner*, may expend resources to increase the informativeness of this private signal, for example by generating a review that is highly tailored to the user’s preferences. This expenditure abstracts the costs of market research, user surveys, and data acquisition required to produce effectively personalized information.

A user’s action, however, is not determined solely by the current private signal. The user also observes predecessor actions—for example, if the previous ten users patronized the restaurant, the next user infers that their private information was likely favorable. This mechanism, known as *social learning*, is a central determinant of opinion formation in domains such as vaccine adoption (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), technology adoption (e.g. Gillingham & Bollinger (2021); Weber (2012)), and political or moral belief formation (e.g., Brady et al. (2021); Guilbeault et al. (2018)). The main object of analysis in this paper is the strategic interaction between a planner that controls private-information precision and agents whose actions are generated by social learning. Consequently, the planner must internalize not only the contemporaneous effect of a signal on the current agent but also the informational *externality* imposed on all subsequent agents through the induced public belief.

Although deployed systems involve many additional institutional and behavioral details, we isolate the relevant mechanism in a tractable model of *controlled social learning*. In the model, an algorithmic planner chooses, at a cost, the precision of the private signal received by each agent in a sequence. The planner does not falsify the latent state or selectively remove observations; instead, its control variable is the degree to which the signal is clear and informative. The analysis therefore asks how a planner with this limited intervention channel can steer collective beliefs, how the optimal intervention varies with the endogenous public belief, and how these interventions affect aggregate welfare.

The solution structure depends on the planner’s objective. An *altruistic* planner, such as an idealized recommendation system, maximizes user utility by increasing the probability that agents choose the state-matching action: patronize the restaurant if and only if it is good. A *biased* planner, such as a system compensated for referrals, instead maximizes the probability of a fixed target action, namely patronage, independently of the true state.

This objective distinction is related to, but not identical with, alignment. By construction, the altruistic planner is aligned with the agent’s *expected utility* under state uncertainty. The biased planner’s objective is state-independent and therefore not generally aligned with expected utility; nevertheless, its intervention is not necessarily welfare-reducing in every realized state. If the latent state supports the target action, as when the restaurant is in fact good, the planner’s induced action coincides with the agent’s *realized utility* optimum. We therefore analyze both aligned and conflicting regimes for altruistic and biased planners.

Contributions and Outline The paper makes the following technical contributions:

1. **A Formal Framework for Controlled Social Learning.** We specify a model that couples a dynamic control problem for a centralized information planner with sequential social learning by decentralized agents. The planner selects the precision of each agent’s private signal to optimize either an altruistic objective, defined by social welfare, or a biased objective, defined by inducing a target action.
2. **Characterization of Optimal Planner Policies.** We characterize optimal policies for altruistic and biased planners as functions of the endogenous public belief. In the altruistic case, the characterization relies on a convexity result for the value function. The resulting regimes specify how precision investment trades off current action incentives against future informational externalities.
3. **LLM-Based Simulation and Strategic Validation.** We implement simulations in which LLMs instantiate both the planner and the agents. The experiments show that (a) a planner that conditions on social learning can shift public opinion and welfare substantially more than a myopic planner, (b) the LLM planner’s emergent strategy broadly follows the theoretically predicted regimes despite non-Bayesian agent behavior, and (c) LLMs can perform strategic reasoning over social-learning dynamics, which creates both capability and risk considerations.

Section 2 positions the contribution relative to prior work. Section 3 defines the model, Sections 4 and 5 derive the optimal policy structure, Section 6 evaluates the theory using LLM-based simulations, and Section 7 concludes. Proofs, derivations, and auxiliary experiments appear in the appendices.

2 RELATED WORK

The analysis is situated at the intersection of social learning, information design, online persuasion and reinforcement-learning formulations, and LLM-based strategic interaction.

Social Learning Bikhchandani et al. (1992) and Banerjee (1992) introduced the canonical sequential social-learning model in economics, and subsequent work has characterized the induced belief dynamics under more general informational and behavioral primitives (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). Related variants modify the social information observed by agents or introduce biases and misperceptions in agents’ inference rules (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010), as well as Bikhchandani et al. (2024) and the references therein).

Work on the *control* of social learning is more limited (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The closest points of comparison are Wei & Anastasopoulos (2022) and Smith et al. (2021). Wei & Anastasopoulos (2022) permits two-way communication in which agents report private signals to the planner in exchange for recommendations. Smith et al. (2021) studies a benevolent planner that directly modifies agents’ choice rules and derives a decentralized pivot mechanism for improved learning. Our formulation excludes signal reports to the planner and leaves action choice decentralized, which better matches information-mediating algorithms that are opaque or not directly perceived by users.

Information Design In our model, the planner optimizes its objective by selecting the precision of agents’ private signals, yielding a constrained instance of classical information design (Bergemann & Morris, 2019). Bayesian persuasion, introduced by Kamenica & Gentzkow (2011), lets a sender choose a signal structure that informs a rational Bayesian receiver about an unknown state. Here, each period contains a costly and restricted information-design subproblem in which the feasible signal is binary and symmetric. The main departure from standard formulations is that receivers are linked by social learning, so each signal design also affects future public beliefs.

Only a small set of papers studies information design in sequential social-learning environments (Arieli et al., 2022; Wu et al., 2025). These works use a one-shot design stage: the sender chooses an information structure at the outset, and the same structure applies to the full sequence of agents. Arieli et al. (2022) studies a self-interested sender with a general Bayesian-persuasion signal structure, whereas Wu et al. (2025) restricts attention to Blackwell-more-informative structures and maximizes the probability of a correct cascade. We instead analyze a dynamic design problem in which the sender chooses a potentially different structure for each agent, which substantially enlarges the strategic policy class.

Online Persuasion and RL Recent work also connects online learning with Bayesian persuasion and information design. In online sequential persuasion, a sender repeatedly interacts with short-lived receivers while the prior over states and/or receiver utilities are initially unknown (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A related line studies sequential persuasion in dynamic environments or MDPs, where state transitions or receiver foresight induce history-dependent and promise-based signaling policies, often with reinforcement-learning interpretations (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). The principal technical objectives in these models are learning guarantees, including sub-linear regret relative to hindsight benchmarks, and approximation guarantees under computational constraints.

By contrast, our planner does not learn unknown environmental parameters. Sequential dependence is generated by social learning among receivers rather than by sender-side parameter learning. The planner also has no informational advantage: it observes neither the latent state nor private signals. Under these restrictions, we obtain structural characterizations of optimal policies. Although the model is sequential, it is not an online-learning or reinforcement-learning environment in the usual sense: planner actions do not change the payoff-relevant state, there are no reward externalities, and

the planner does not explore to estimate unknown parameters. The planner’s information changes only through observed actions, which encode belief propagation through social learning. Thus, the relevant intertemporal channel is an information externality from current actions to future public beliefs.

LLMs A newer literature studies LLMs as components of information-design systems (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These papers focus on one-to-one interactions and therefore do not model the social dynamics central to our setting. The closest work is Duetting et al. (2025), which allows a sender to choose both a signal structure and a framing that affects the receiver’s posterior, including under non-Bayesian response rules. Their use of an LLM-based oracle addresses the computational difficulty of optimizing over the framing space and builds on evidence that LLMs can approximate humans as economic agents (Dillion et al., 2023; Horton, 2023). We do not study framing or isolated receiver interaction; instead, we adapt parts of their empirical methodology to evaluate repeated social-learning dynamics in Section 6.

3 SOCIAL LEARNING MODEL

The model consists of a planner and a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature draws a fixed but *unknown* exogenous state $\omega \in \Omega := \{G, B\}$ with commonly known prior $\mathbb{P}(\omega = G) = b_1$. In the recommendation-system example of Section 1, state G denotes that the restaurant is good.

At each time i , the planner chooses the level of personalization, equivalently the precision, for agent i ’s information. The agent then receives a private signal $s_i \in \Omega$. The signal is binary and symmetric: it equals the latent state ω with probability $q_i \in [0.5, 1]$, i.e., $\mathbb{P}(s_i = \omega) = q_i$. We refer to q_i as agent i ’s signal precision.

Conditional on ω , the private signals are independent across agents and independent of the public history. Agent i observes her signal and the actions of all predecessors $j < i$, then selects an action $a_i \in \Omega$. Payoff is normalized to 0 when $a_i = \omega$ and to $-C$ otherwise, with $C > 0$; hence each agent chooses an expected-utility-maximizing action. In the running example, this utility specification corresponds to patronizing the business exactly when the state is G .

3.1 AGENTS’ DECISION PROBLEMS

Each agent observes the complete sequence of predecessor actions together with their signal precisions. We denote this public history by $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because agents and planner share exactly this information, they also share a *public* posterior belief about ω , updated after each action. The public belief immediately before agent i acts is $b_i = \mathbb{P}(\omega = G | \mathcal{H}_i)$, with initial value b_1 equal to the common prior over Ω . As in Banerjee (1992) and Bikhchandani et al. (1992), the process $(b_i)_{i \in \mathbb{N}}$ is Markov and b_i is a sufficient statistic for $\mathcal{H}_i$ (see Appendix B.3 for proof).

Given $\mathcal{H}_i$ (equivalently, by sufficiency, b_i) and q_i , agent i chooses $a_i \in \Omega$ to maximize expected utility. Ties are resolved in favor of the action matching s_i . Combining the public belief with the private signal of precision q_i , the agent forms the following *private* posterior belief $\tilde{b}_i$ about ω (derived in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

The decision rule selects the state with larger posterior probability: $a_i = G$ if $\tilde{b}_i > 0.5$, $a_i = B$ if $\tilde{b}_i < 0.5$, and $a_i = s_i$ under equality. Substituting the posterior expression yields the threshold rule

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Subsequent agents observe a_i , so the public belief is updated to b_{i+1} according to the following recursion (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. When $1 - q_i \leq b_i \leq q_i$, Equation (2) implies that agent i 's action reveals her private signal exactly; the updated public belief therefore equals that agent's private posterior. Outside this interval, Equation (2) implies that the private signal does not affect the action, so the action contains no additional information and the public belief remains fixed. The process then enters an absorbing state, called an information cascade or herding state in the classical social-learning literature (Bikhchandani et al., 1992; Banerjee, 1992). In such states, public belief no longer incorporates agents' private information.

Agent i 's expected utility is $-C \mathbb{P}(a_i \neq \omega | b_i, q_i)$, which reduces to the following expression (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We analyze two planner objectives. An altruistic planner seeks to induce the state-matching action ($a_i = \omega$), whereas a biased planner seeks to induce the fixed action G independently of ω . We use subscripts A and B for the altruistic and biased planners, respectively. In both cases, the control variable is the precision of each agent's private signal. The cost of precision is represented by a function $\beta(\cdot)$ that is non-negative, increasing, continuous, and concave in its argument. The planner's information set is exactly $\mathcal{H}_i$, identical to the agents' public information.

Altruistic Planner At time i , the altruistic planner chooses precision q_i for agent i and pays $\beta(q_i)$, where $\beta(p) = 0$ for a baseline precision $p \in [0.5, 1]$. For the altruistic planner, lowering precision is weakly dominated (see Appendix C.9). We therefore normalize costs so that only precision increases above p generate additional cost, with total cost increasing and marginal cost decreasing in the chosen precision. Agents know both p and $\beta(\cdot)$.

The altruistic planner maximizes expected social welfare net of precision-investment costs, where social welfare is the expected aggregate utility of the agents. Let $r_A(b_i, q_i)$ denote the one-period reward when the public belief is b_i and the planner chooses precision q_i for agent i . Starting from public belief b_1 and using a policy sequence $\pi = (\pi_i)_{i=1}^\infty$ with $\pi_i(\mathcal{H}_i) = q_i$, the planner's discounted objective with discount factor $\delta \in [0, 1]$ is

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The altruistic optimal value and optimal policy are the supremum and corresponding arg supremum of $V_A^\pi(\cdot)$ over $\pi \in \Pi$, where Π denotes the feasible policy class. The myopic formulation is the special case $\delta = 0$.

Biased Planner The biased planner differs from the altruistic planner in objective and therefore in its reward and cost terms; Appendix A gives a political-campaign interpretation. The biased planner seeks to induce action G from every agent, independently of ω . If the agent chooses G the planner incurs cost 0, whereas if the agent chooses B the planner incurs cost $C > 0$, again independently of ω . Unlike the altruistic planner, the biased planner may increase or decrease private-signal precision relative to the baseline p , and either deviation is costly. Any precision $q_i \neq p$ requires tailoring the message to the agent, which in the application corresponds to learning how that agent processes content. The biased planner therefore pays $\beta(|q_i - p|)$ when choosing precision q_i , with $\beta(0) = 0$.

The biased planner's expected one-period reward at time i is therefore

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

Given Equation (6), the biased-planner quantities $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ are defined analogously to $V_A^\pi(\cdot)$, $V_A^*(\cdot)$, and $\pi_A^*(\cdot)$ under Equation (5).

As specified in Section 3, each agent obtains higher utility from an action matching ω . Hence, when $\omega = B$, successful persuasion by the biased planner lowers the agent’s realized utility. By contrast, successful intervention by the altruistic planner always increases the probability of a state-matching action. The altruistic objective is therefore aligned with each agent’s utility for both states, whereas the biased objective is aligned only when $\omega = G$.

For both objectives, the planner’s problem is an infinite-horizon discounted stationary Markov Decision Process with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition kernel induced by Equation (3) (Puterman, 1990). A unique optimal value function therefore exists (Kallenberg, 2011). We restrict attention to deterministic Markov policies, so $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$.

Remark 2. *The model imposes three substantive assumptions. (1) The planner and agents observe the same public history. This assumption excludes planners with richer private data, but it is appropriate when all parties condition on the same public evidence, such as early findings or survey data. (2) The planner controls signal precision while the signal remains a binary symmetric channel. This assumption excludes cherry-picking, censorship, and falsification, but it captures mediators that only improve or degrade the quality of noisy information sources without changing their content. (3) The planner’s control choices are fully observable. This assumption excludes covert policies and hidden framing, but it matches transparent mediation environments.*

4 OPTIMAL ALTRUISTIC POLICIES

We first analyze the myopic benchmark, which removes the intertemporal value of social learning. Formally, this benchmark sets the discount factor to $\delta = 0$, so the planner optimizes only the current-period reward. Under $\delta = 0$, $V_A^\pi(b) = r_A(b, \pi(b))$, and the optimal myopic policy $\pi_A^0(\cdot)$ satisfies

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

Equivalently, the myopic altruistic problem can be interpreted as a decentralized problem in which each agent chooses her own signal precision and pays the associated cost to maximize her own expected utility net of that cost.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimum has a threshold structure. When the public belief is sufficiently extreme, the planner selects the zero-cost baseline precision p . When the public belief is sufficiently diffuse, the planner selects a perfectly informative signal, i.e., precision 1. The threshold depends only on the cost of perfect precision, $\beta(1)$, relative to the loss from an incorrect action, C . If $\beta(1) \geq C$, perfect precision is never cost-effective; consequently $t_M = 0.5$, and the belief interval on which myopic optimal precision equals 1 is empty.

The following structural property of the altruistic value function is central to the dynamic characterization.

Theorem 2. *$V_A^*(\cdot)$ is convex with respect to public belief.*

The proof of Theorem 2, given in Appendix C.3, is technically involved and may be independently useful. The main difficulty is that agents’ action rules depend endogenously on the public belief. If actions were exogenous to the public-belief process, as in settings such as Nyarko (1994), expected utility would be linear in the belief state and value-function convexity would follow directly.

Convexity provides the key regularity condition used to characterize the optimal policy.

Theorem 3. *There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and*

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & \text{o.w.} \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal policy partitions the public-belief space into three regimes. First, as in the myopic benchmark, the dynamic optimum does not invest in additional precision when the public belief is sufficiently extreme. However, this no-investment region requires a more extreme belief than in the myopic case because the dynamic planner internalizes the informational effect on future agents.

When the public belief is close to 0.5 and therefore weakly informative, the dynamic optimum selects precision 1 whenever the cost is not prohibitive. Under Equation (2), the agent's action then equals the true state with probability 1, and the public belief updates immediately to either 0 or 1.

In the intermediate regime, the dynamic optimum chooses the minimal precision $\max(b, 1 - b)$ that makes the agent's action depend on her private signal, as specified by Equation (2). Any lower precision leaves the action independent of the private signal and hence uninformative relative to the existing public belief. This regime therefore implements the lowest-cost precision that preserves social learning from observed peer actions.

5 OPTIMAL BIASED POLICIES

We again begin with the myopic benchmark, now for the biased planner. Analogously to Section 4, and following the same construction used in Section 4, the myopic biased policy $\pi_B^0(\cdot)$ is defined as the arg supremum of the one-period reward $r_B(b, q)$. The following theorem characterizes $\pi_B^0(\cdot)$. Unlike in the altruistic case, an exact maximizer need not exist for all beliefs (Puterman, 1990, sec. 2.3.1). Where the supremum is not attained, we use ϵ -optimal policies, indicated by an ϵ superscript.

Theorem 4. *There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:*

- (A) *If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.*
- (B) *If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.*
- (C) *If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.*
- (D) *If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

The proof appears in Appendix C.6. We next use Theorem 4 to obtain the dynamic biased-policy characterization.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) *If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.*
- (B) *If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.*
- (C) *If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.*
- (D) *If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.*
- (E) *If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon*}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Theorem 5, proved in Appendix C.8 and illustrated in figure 2a, partitions beliefs into five policy regimes.

In regime (A), the public belief is sufficiently extreme that the cost of steering the process can exceed the expected benefit, including when $b < 0.5$ and the implication for planner utility is adverse. The chosen precision p is below $\max(b, 1 - b)$; by Equation (2), the agent then follows the public belief regardless of the private signal. The action is uninformative, the public belief remains fixed, and the Markov policy repeats this behavior indefinitely. The resulting cascade is unfavorable for the planner when $b < 0.5$ and favorable otherwise.

In regime (B), the public belief is near an unfavorable cascade. The planner raises signal precision enough for the private signal to affect the action, even though the signal is more likely to be B in expectation. This intervention creates a positive probability that the agent takes the planner-favorable action, functioning as a costly attempt to move the process away from the unfavorable absorbing region.

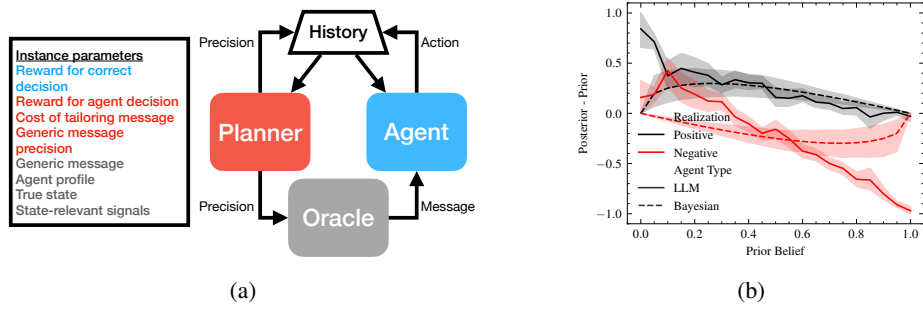


Figure 1: (a) Experimental system diagram. Instance parameters are color-coded by the LLM role that uses them. (b) Belief changes for LLM agents (solid) and Bayesian agents (dashed), evaluated at a fixed prior after either a positive (black) or negative (red) signal.

In regime (C), for slightly larger beliefs with $b < 0.5$, the planner may reduce precision below p . Because higher precision is more likely to communicate unfavorable information in this region, the planner selects the least precise signal that can still affect the action, namely a precision satisfying $q \geq 1 - b$, to preserve a chance of moving the public belief in a favorable direction.

In regime (D), the public belief already weakly favors the target action. Since $b > 0.5$, increasing precision raises the probability that the agent infers $\omega = G$. The planner therefore chooses precision at least p to reinforce the favorable public belief.

In regime (E), the public belief is sufficiently favorable that the planner reduces precision just below $\max(b, 1 - b)$, specifically to $b - \epsilon$. The agent then ignores the private signal and chooses G . In this region, preventing a contrary private signal from overturning favorable public belief is more valuable than both the cost of lowering precision and the possible benefit from further belief improvement.

6 EVALUATION VIA LLM-BASED SIMULATION

We now utilize LLMs to empirically study the implications and applicability of our theoretical results and model. Our evaluation proceeds in three steps: we first analyze the behavior of LLM agents to identify key deviations from Bayesian rationality, then examine how an LLM planner adapts its strategy in response, and finally evaluate the resulting impact on social welfare.

To instantiate the controlled social-learning model empirically, we use a car-purchase scenario in which agents sequentially decide whether to buy a new car. As shown in figure 1a, the simulation assigns LLMs to three functional roles, with the corresponding prompts reported in Appendix E.2.2:

Agent: Conditional on an assigned profile, the agent observes previous actions and a private car-related message of known precision, reports a belief about car quality, and chooses whether to buy.

Planner: Conditional on the action history, the planner selects the private-signal precision q_i for agent i according to its objective.

Oracle: Conditional on the planner’s precision choice, an agent profile, and the car fact sheet (see Appendix E.2.1), the oracle generates a tailored private signal with the requested precision. We compare the LLM-simulation outcomes with numerical solutions of the planner’s MDP. In both experimental modalities, the cost function is linear, $\beta(q) = k|q - p|$; we vary k , the baseline precision p , and the discount factor δ , while fixing the incorrect-action cost at $C = 1$. Appendix E provides the experimental and prompting details, and Appendix E.3 validates the belief reports and oracle precision generation.

The experiments support three conclusions:

1. **Control of social learning is consequential:** Both analytical and LLM planners substantially improve planner utility when they condition on social-learning dynamics. The welfare effect is positive or negative depending on the alignment between planner and agent objectives.

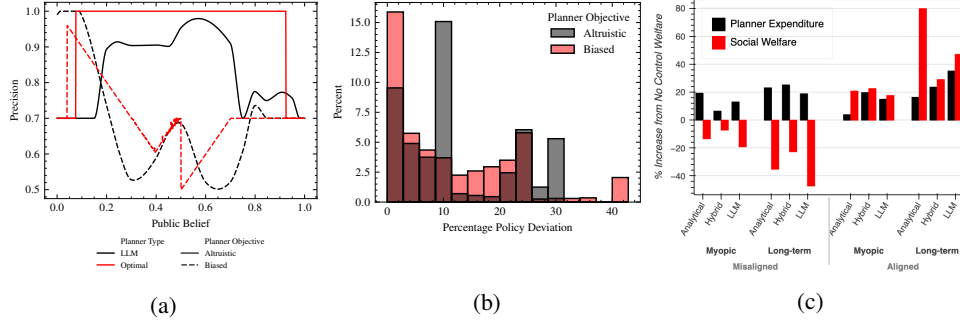


Figure 2: (a) Representative policies for the LLM planner (black) and analytically optimal planner (red), shown for altruistic (solid) and biased (dashed) objectives. (b) Distribution of percentage deviations between LLM and optimal policies. (c) Planner expenditure and social-welfare change, each reported as a percentage of the no-control baseline welfare. The true state is fixed to B; the left panel corresponds to a biased planner targeting action G, and the right panel corresponds to an altruistic planner.

2. **The analytical characterization is structurally robust:** LLM planners select policies that are close to the non-obvious analytically optimal policies, despite interacting with non-Bayesian LLM agents.
3. **LLMs exhibit emergent strategic adaptation:** LLM planners display strategic structure in their policy choices. Their policies are close to the analytical optima and deviate in directions consistent with adaptation to the non-Bayesian behavior of the LLM agents they face.

6.1 LLMs AS NON-BAYESIAN AGENTS

The theoretical analysis assumes Bayesian agents, whereas both humans and LLMs exhibit systematic cognitive biases (Rupprecht et al., 2025). To isolate the relevant LLM deviations, we first measure how an LLM updates a reported belief after receiving new information. The results in figure 1b identify the following non-Bayesian updating patterns (see also Appendix E.4):

- (NB1) LLM agents underreact to private signals which align with their prior beliefs.
- (NB2) LLM agents overreact to private signals which run counter to their prior beliefs.
- (NB3) As a result, LLM agents require a stronger public belief to enter an information cascade.

The same qualitative biases have been documented in human experiments (e.g., Ba et al. (2022); Chan et al. (2025)). The subsequent results show that the LLM planner’s policy remains structurally close to the analytical characterization even when agents exhibit these non-Bayesian updates.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We compare LLM-planner policies with the optimal policies derived in Sections 4 and 5. As shown in figure 2a, the LLM planner’s emergent strategies remain structurally similar to the theoretical optima despite non-Bayesian agent behavior. Thus, even for a highly nontrivial optimum such as the biased policy in figure 2a, the LLM planner exhibits strategic behavior that conditions on and exploits social-learning dynamics.

For the altruistic objective, both planners stop investing when the public belief is strong and invest heavily when it is near the uncertain region. For the biased objective, both planners display the same qualitative regimes: high investment near an unfavorable belief, reduced precision around the midpoint, and no investment once a favorable belief is established. As shown in figure 2b, policy deviations are often small; for both planner objectives, the deviation is below 10% for a majority of belief states. This supports broad structural alignment between the LLM strategy and the theoretical characterization.

The remaining structural differences are consistent with strategic adaptation to the non-Bayesian behaviors identified in Section 6.1. (1) The LLM planner tends to avoid boundary precisions, 0.5 and 1.0, consistent with central-tendency bias (Rupprecht et al., 2025). (2) The LLM policy tapers investment gradually rather than imposing a sharp cutoff. This pattern is consistent with agents’ resistance to cascades (NB3): because agents with strong priors may still respond to countervailing private signals, stopping investment is less safe than in the Bayesian model. (3) The biased LLM planner also continues investing at very low beliefs, consistent with anticipating that agents may overreact (NB2) to an unexpectedly positive signal; consequently, a last-resort intervention remains more valuable than under Bayesian updating.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as agents’ total expected discounted utility. Because more informative signals weakly improve an agent’s expected utility, welfare is monotone in signal precision (see Appendix C.9); hence the altruistic planner weakly increases welfare relative to the baseline. To quantify these effects, we compare welfare and planner expenditure in three regimes: (1) the *analytical* regime, with the optimal policy and Bayesian agents; (2) the *LLM* regime, with an LLM planner and LLM agents; and (3) the *hybrid* regime, with the optimal policy applied to LLM agents.

The results in figure 2c show that planners in all regimes can materially change social welfare. Myopic policies, which ignore social learning, also produce substantially worse planner outcomes. When misaligned, the biased analytical and LLM planners reduce social welfare by 40 to 50%. This reduction is achieved through policies that intentionally lower signal precision and thereby obscure information about the true state. The welfare loss is notable given the stringent planner constraints in Remark 2; it therefore provides evidence of the risk posed by misaligned LLM information mediators.

The results also quantify the cost of behavioral misspecification. In the hybrid regime, the analytically optimal policy is designed for Bayes-rational agents and performs worse when deployed against non-Bayesian LLM agents. The LLM policy, by contrast, remains close to the analytical policy while better matching agents with human-like updating biases.

7 CONCLUSION

We introduced a formal framework for strategic control of social learning by an algorithmic information mediator. The analysis shows that even a planner restricted to controlling signal precision can substantially affect social welfare in either direction, and that effective intervention requires accounting for the informational externalities generated by social learning. The LLM simulations support these conclusions: LLM planners exhibit strategic reasoning that matches the qualitative structure of the analytical policies while adapting to non-Bayesian LLM agents.

These findings imply three technical and policy-relevant lessons for algorithmic information mediation. First, human-AI and human-algorithm interactions should not be modeled in isolation when their outputs interact with social learning; otherwise, the analysis can miss both beneficial and harmful amplification channels. Second, information mediators can substantially redirect social learning and public opinion even through the comparatively narrow intervention of precision control, which motivates formal study of mitigation mechanisms. Third, modern LLMs can implement strategies that exploit both social-learning dynamics and non-Bayesian updating biases. A limitation of the present study is the absence of human-subject data. Because the fidelity of LLM-based human simulators remains contested (see, e.g., Horton (2023); Gao et al. (2025)), human data would be useful for validating, calibrating, or fine-tuning the LLM simulators.

Two directions are particularly important for future work: generalizing the model and mitigating welfare losses from biased planners. Several extensions would relax core assumptions, including larger state spaces, richer signal structures, more general loss functions, and heterogeneous agents. Moving beyond binary states and 0-1 losses appears primarily algebraic, and we conjecture that the qualitative policy regimes persist. For richer signal structures, the central technical obstacle is preserving the value-function convexity property used in the dynamic characterization. Identifying the broadest class of signal structures that maintains this property is a natural open problem. As

an initial heterogeneous-agent extension, Appendix D generalizes the altruistic-planner results to agents with opposing preferences.

A second direction is the design of interventions that prevent welfare degradation from biased planners. This includes regulatory or mechanism-design approaches that align planner incentives with agent welfare and thereby reduce harmful information mediation.

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A APPLICATIONS ELABORATED

Consider a sequence of individuals, modeled as agents, who decide whether to patronize a business or service provider such as a restaurant, contractor, or realtor. A recommendation system, modeled as the planner, provides information about the business. The binary latent state indicates whether the business is good. Both the recommendation system and the agents prefer actions that match this latent state: agents should patronize the business if and only if it is good, and the recommendation system seeks to induce exactly those actions. Under this objective, the recommendation system is an *altruistic planner*.

The recommendation system may provide a highly targeted advertisement, which corresponds to a *precise* signal, that presents the business’s strengths and weaknesses in a format matched to the agent’s background and characteristics. Alternatively, it may provide a generic or confusing advertisement that has little decision value. A more precise signal is more informative, more comprehensible, and better adapted to the agent, so it increases the probability of the correct action: patronage if and only if the business is good. Because social welfare increases with correct decisions, precision is beneficial; however, modifying precision is costly because it requires tailoring to agent-specific attributes. The recommendation system therefore chooses precisions to maximize social welfare net of tailoring costs.

As an example of a *biased planner*, consider voters who prefer to support the candidate most likely to win in their community. Such “back-a-winner” incentives affect electoral outcomes and help explain the disproportionate influence of early US primary states (see Bartels (1988)). The planner is a particular candidate’s campaign, whose objective is to induce support for that candidate. The binary latent state indicates whether the candidate is winning. If a voter observed the true state, she would support the candidate if and only if he were winning.

Each agent has local knowledge of her community but can infer whether the candidate is winning only if she accurately understands the candidate’s stances and policies. Accurate understanding depends on the format in which information is delivered. Some agents may process audio-visual material most effectively; others may require long-form articles with facts, figures, and citations; others may respond best to brief focused content or to content in a native language.

The campaign sends digital advertisements, interpreted as private signals, with varying precision. Precision measures how well the content format is tailored to the agent. A precise signal accurately communicates the relevant state by enabling the agent to understand the candidate's policies, record, and character, thereby allowing her to infer whether the candidate will win. Precision does not always benefit the campaign: if the candidate is losing in the agent's community, a highly precise signal would induce opposition. An imprecise signal may instead make support more likely by obscuring the true state, for example through vague or confusing content. The biased planner may therefore optimally reduce precision when it expects the candidate to be losing. Such reduction is still costly because it requires knowledge of how to tailor confusion to the specific agent; for instance, the planner may need to know that an agent who prefers concise messages can be confused by verbose long-form content. The planner consequently chooses precisions to maximize the expected number of supporters net of message-generation costs.

B SOCIAL LEARNING DYNAMICS

At each step i , the timing is as follows:

1. The planner selects q_i as a function of $\mathcal{H}_i$, and all agents observe q_i .
2. Agent i receives private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i chooses a_i from the observable history and private signal to maximize expected utility.
4. All subsequent agents and the planner observe a_i , and the public belief is updated to b_{i+1} .

Because q_i is a function of $\mathcal{H}_i$, conditioning on both q_i and $\mathcal{H}_i$ is equivalent to conditioning on $\mathcal{H}_i$ alone. We use this redundancy throughout the derivations.

B.1 PRIVATE BELIEF UPDATE

We next derive agent i 's posterior and action rule. Let $\alpha_i := \frac{b_i}{1-b_i}$ denote the public likelihood ratio implied by observed actions. After observing s_i , agent i forms the private posterior $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$ and the associated private likelihood ratio $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. The following cases compute $\tilde{\alpha}_i$ and $\tilde{b}_i$.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \tag{8}$$

$$= \frac{q_i}{1-q_i} \alpha_i \tag{9}$$

In obtaining Equation (8), we apply the assumption that, conditioned upon ω , agents' signals are independent of one another and the history to assert that $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$. The same is done when conditioning upon $\omega = B$. We additionally use the fact that q_i is a function $\mathcal{H}_i$.

2. If private signal is $s_i = B$, following the logic above,

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \tag{10}$$

Let $y(b, q)$ and $z(b, q)$ denote the conditional probabilities of observing signals G and B , respectively, given public belief b and precision q . Conditioning on ω and applying the law of total probability gives

$$\begin{aligned} y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\ &= q_i b_i + (1-q_i)(1-b_i) = 1 + 2q_i b_i - q_i - b_i \end{aligned} \tag{11}$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \tag{12}$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1-q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \quad (13)$$

If $s_i = G$, then $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i) > 0$ on the realized event; analogously, if $s_i = B$, then $z(b_i, q_i) > 0$. The posterior expression is therefore well-defined.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i chooses G when $\tilde{b}_i > 0.5$ (equivalently, $\tilde{\alpha}_i > 1$), chooses B when $\tilde{b}_i < 0.5$ (equivalently, $\tilde{\alpha}_i < 1$), and chooses s_i under the tie-breaking rule. The action and precision q_i are publicly observed and determine the next public belief.

If $b_i < 1 - q_i$, then $\alpha_i < \frac{1-q_i}{q_i}$. Hence $s_i = G$ implies $\tilde{\alpha}_i < 1$ by Equation (9), and $s_i = B$ implies $\tilde{\alpha}_i < \frac{(1-q_i)^2}{q_i^2} \leq 1$ by Equation (10), since $q_i \geq 0.5$. Thus $a_i = B$ for every signal realization. A symmetric argument shows that $a_i = G$ when $b_i > q_i$. In both cases, the public belief is sufficiently strong to dominate the private signal.

Now suppose $1 - q_i \leq b_i \leq q_i$. If $s_i = G$, then $\tilde{\alpha}_i \geq 1$ by Equation (8) and $b_i \geq 1 - q_i$; if $s_i = B$, then $\tilde{\alpha}_i \leq 1$ by Equation (10) and $b_i \leq q_i$. With the tie-breaking rule, the action equals the signal, $a_i = s_i$. This yields Equation (2), reproduced here.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Therefore, the action is informative only in the first case of Equation (2), where future agents can infer agent i 's private signal exactly. In that case, the public belief updates as in the private posterior Equation (13). In the remaining cases of Equation (2), the action is independent of the private signal and contains no additional information about the state, so the public belief is unchanged. This gives Equation (3), reproduced here:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (14)$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now show that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ depends on the history only through b_i and q_i , for each $a \in \{G, B\}$. Hence

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

From Equation (2), for the second and third cases, it is clear that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, regardless of the rest of $\mathcal{H}_i$. We consider the first case next.

$$\begin{aligned} & \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) \\ &= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first equality applies the law of total probability. The second uses that q_i is a function of $\mathcal{H}_i$, so $\mathbb{P}(\omega = w | \mathcal{H}_i, q_i)$ equals the public belief term. The third uses the condition $1 - q_i \leq b_i \leq q_i$, under which $a_i = s_i$, together with conditional independence of s_i from the history given ω .

This establishes the following Markovianity statement.

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G | b_i, q_i, a_i\} \quad (17)$$

Equivalently, b_i is a sufficient statistic for all information in H_i relevant to updating b_{i+1} .

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G | b_i, q_i\}}{\mathbb{P}\{a_i | b_i, q_i\}}$$

We will prove the claim via induction on i .

For our base case, consider $i = 1$ so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem provides the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G | b_1, q_1\} \mathbb{P}\{a_1 | \omega = G, b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1 | \omega = G, b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G | b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\} \mathbb{P}\{\omega = G | b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G | b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} \end{aligned}$$

where the final step follows as the unconditioned belief of ω is equal to the prior belief b_1 .

Now, assume the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G | \mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_n, q_n\} \mathbb{P}\{a_n | \omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | \mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n | \omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | \mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G | \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | b_n, q_n\} \mathbb{P}\{\omega = G | \mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G | b_n, q_n\}}{\mathbb{P}\{a_n | b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n .

This completes the proof. $\square$

The argument adapts the Markovianity proofs of Banerjee (1992) and Bikhchandani et al. (1992). The additional steps account for the planner's endogenous selection of agents' signal precisions.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We verify the expression separately for each action regime in Equation (2). In the second regime,

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \mathbb{P}(\omega \neq G | b_i, q_i) = 1 - b_i$$

In the third regime of Equation (2), analogously,

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \mathbb{P}(\omega \neq B | b_i, q_i) = b_i$$

Finally, when $1 - q_i \leq b_i \leq q_i$, Equation (2) gives $a_i = s_i$, so

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqn 2} \quad (18)$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqns 11, 12, Thm 6} \quad (19)$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1 - q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \quad \text{Eqn 13} \quad (20)$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three regimes proves the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) gives the derivative structure

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As stated in Section 3.2, the altruistic planner can be restricted to precisions $q \in [p, 1]$ because social welfare is increasing in precision (see Appendix C.9). Under this normalization, $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) is optimized at precision p whenever $p < \max(b, 1 - b)$. Thus, if the signal precision is too low to affect the agent's action, the planner does not spend resources increasing it. In the latter case, $\beta(\cdot)$ is increasing and concave, so $\dot{\beta}(\cdot)$ is positive and decreasing. Therefore, if the optimum lies in this region, it is attained at $q = 1$.

It remains to identify when the latter region dominates the baseline choice.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, the inequality is never satisfied, so precision 1 is never optimal. Otherwise, the inequality yields $b \in [\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C}]$, which proves Theorem 1. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We prove convexity by induction on the expected k -th stage reward, defined as the planner's expected utility from the control and action associated with the k -th agent. The base case is the instantaneous reward in Equation (5), which is convex in public belief.

The main technical issue is the form of the public-belief update. Although the belief state space is uncountable, each update has support on at most two successor beliefs. We exploit this structure by defining a decision tree: the complete binary tree of possible trajectories after fixing an initial belief and policy. Each node has two children, corresponding to the two possible signal realizations for the next agent. The root represents the expected instantaneous reward at time 1, and level k contains 2^{k-1} nodes indexed by sequences of $k - 1$ signal realizations.

The induction step shows that if a node at level $k - 1$ has convex expected reward, then its two children at level k preserve the required convexity property. The obstacle is that agents' actions depend on public

belief: under the same precision and signal realization, agents with different public beliefs may take different actions (see Equation (2)). Consequently, standard Markov-process bounds on continuation values do not apply directly. The specific Bayesian update structure resolves this difficulty because public beliefs form a martingale, i.e., $\mathbb{E}[b_{i+1}] = b_i$. Combining this martingale property with instantaneous-reward convexity completes the induction and the proof.

Preliminaries We first establish several basic properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1-b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward r_A . The action rule in Equation (2) is symmetric, so $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Therefore,

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Hence the instantaneous reward function is symmetric around 0.5.

Now consider the belief-state update starting from n_i under precision q_i . Let $\bar{x}$ and $\underline{x}$ denote the updated beliefs from x after observing actions G and B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \underline{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus the belief updates are symmetric. Combining this fact with reward symmetry gives

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \underline{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Therefore, the expected rewards and belief-state values in Λ_n and Λ_b are mirror images.

Finally, define the branching probabilities $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Hence the branching probabilities of Λ_n and Λ_b are symmetric, completing the proof.

Extremal values

It suffices to show $V_A^*(0) = 0$, because the symmetry about 0.5 established above implies $V_A^*(1) = 0$.

From Equation (3), when $b_0 = 0$, both successor beliefs satisfy $b_0^+ = b_0^- = 0$ for any precision. Hence, for all signal realizations and controls, actions and public belief remain fixed: $b_i = 0 \forall i$ and, by Equation (2), $a_i = B \forall i$.

Equation (5) then gives

$$\begin{aligned}
V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right]
\end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We now introduce the concept of *decision trees* for Markovian and more general policies. A policy can be represented by a decision tree, which is a complete binary tree as in figure 3. Every node of the decision tree represents the state of the system, i.e., the corresponding belief value when the controller sends its private signal to the agent corresponding to the level of the node. The branches represent the actions taken by the agent corresponding to the level once it receives its private signal. The controller's policies (i.e. choice of precision) determine the probabilities of the actions from each node and the belief resulting from the action. The probabilities of an action at a certain node can be considered the weight of the branch. A decision tree is uniquely identified by the belief values associated with nodes and the weights of the branches. For a deterministic stationary Markovian policy, the controller's choice of precision, and therefore the probability associated with an action, is a deterministic function of the belief value of the node, regardless of the level of the node in the tree and the path to the node from the root. Thus, the decision tree of any such policy, i.e. the belief values at the nodes and the weights of the branches, is determined entirely by the value of the initial state. For an arbitrary policy of the controller, the controller's decision at a given node can depend on the entire path leading to the node.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We will first show that, for fixed precision, the instantaneous reward r_A is convex in public belief. We can rewrite r_A as follows using Lemma 7:

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave. As we are subtracting the minimum, the overall function is convex with respect to x .

We consider arbitrary x_0, m_0, n_0 such that $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. To show the convexity of $V_A^*(\cdot)$ we must show that $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. By definition of the optimal value function, $V_A^*(x) \geq V_A^\pi(x)$ for any policy π . Thus, it will suffice to show that there exist $\tilde{\pi}$ such that

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

We refer to the decision tree of π^* when the initial state is x_0 as Λ .

We consider $\tilde{\pi}$ to be a policy of the controller that chooses the precision at each node to be the same as what Λ does at the counterpart node. The nodes of decision trees are considered to be counterparts if they can be reached via the same sequence of actions from the root (see figure 3 for illustration). Thus, the precision chosen by this policy is the same as what Λ does after the same set of actions of the agents. We refer to the decision trees for $\tilde{\pi}$ with initial state m_0 (respectively, n_0) as Λ_m (Λ_n).

We now illustrate the policy $\tilde{\pi}$ for initial values m_0 and n_0 considering the decision trees Λ_m and Λ_n . While doing so, we illustrate counterpart nodes and the weights of the branches. We will select one path of actions λ out of these decision trees by specifying set $\mathcal{G}$ as the epochs where action G is taken and $\mathcal{B}$ as its complement. Refer to figure 3 for illustration. Considering the decision tree Λ , we will refer to x_i as the state at epoch i when beginning with belief x_0 and following π^* along this path of actions. That is, x_i assumes one of 2^i possible belief values at epoch i starting from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote the

public belief after beginning with m_0 (n_0), following the path λ . Node m_i (respectively n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i from decision tree Λ . Note that, $\tilde{\pi}(m_i)$ can now be illustrated notationally. Specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Similarly, $\tilde{\pi}$ can be defined at each node of Λ_m , considering the action taken by π^* at each counterpart node in Λ . Note that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m . This extends similarly to branches associated with action B and Λ_n . Thus, the three decision trees differ in both the values of the nodes and the weights of the branches. Since π^* is deterministic, so is $\tilde{\pi}$. Therefore, from Equation (13) and Equation (3), x_i , m_i , and n_i are deterministic quantities once the specific path of actions is specified by $\mathcal{G}$.

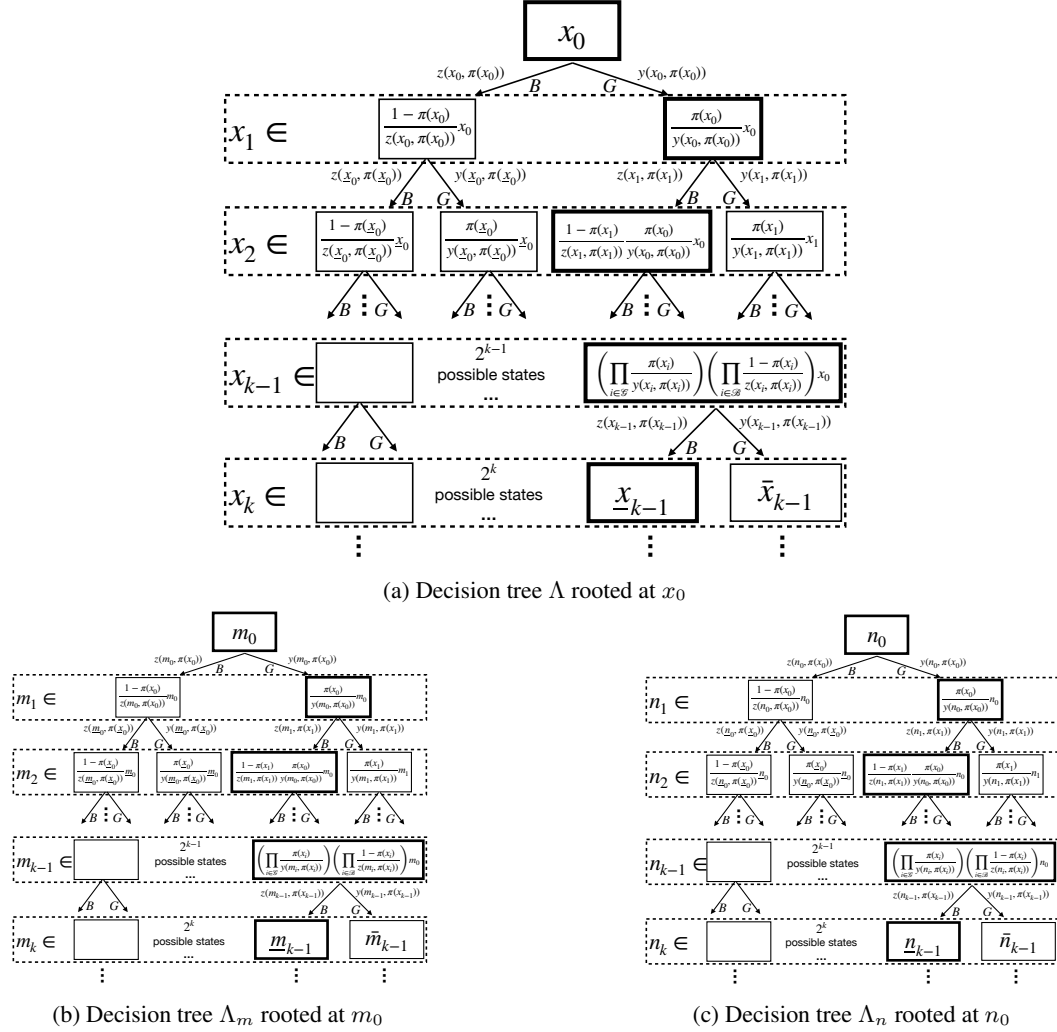


Figure 3: Here we depict the belief evolution in the form of three binary trees. The tree begins with state x_0 , m_0 , and n_0 , respectively, and level i contains 2^i possible states reachable after i epochs. Left branches correspond to action B leading to updated belief $\underline{x}_0$ (respectively, $\underline{m}_0$ and $\underline{n}_0$). Right branches correspond to action G leading to updated belief $\bar{x}_0$ (respectively, $\bar{m}_0$ and $\bar{n}_0$). Each branch is labeled with the corresponding probability, and the bold states indicate an example path, like λ . In the example path, $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. The bold states in each level of the three trees are counterparts.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ be the expected instantaneous reward at epoch k when beginning with state b_0 and applying policy π and $\phi^*(x) = \phi^{\pi^*}(x)$. Applying Fubini's Theorem, we can rewrite our value function

as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus, we can rewrite (24), as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

To prove this, we show that for all $i = 0, 1, 2, \dots$

$$\phi_i^*(x_0) \leq t \phi_i^{\tilde{\pi}}(m_0) + (1-t) \phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

leading directly to (26), (24), and finally the convexity of $V^*(\cdot)$ with respect to public belief.

Now let us consider general $i = k$. $\phi_{k-1}^\pi(x_0)$ will have 2^{k-1} terms each corresponding to one path of possible actions in the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ will yield the following term in $\phi_{k-1}^*(x_0)$

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, that is, the probability of following the specified path λ when beginning with state x_0 and following policy π_A^* . Again, we will use $\bar{x}_i$ and $\underline{x}_i$ to represent updated beliefs from x_i after observing action G or B , respectively. Thus, $\bar{x}_i$ and $\underline{x}_i$ constitute belief values of the children nodes (in the $(i+1)$ th level) of the node with belief value x_i in the i -th level. Refer to levels $k-1$ and k in figure 3a. Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (29)$$

Similarly, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, that is, the probability of following the specified path λ when beginning with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. We also define $P_\lambda^{\tilde{\pi}}(n_0)$ by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (31)$$

We will show that

$$\begin{aligned} & P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \leq t P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & + (1-t) P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \end{aligned} \quad (32)$$

Similarly, each pair of children for each term in the $(k-1)$ -th level of the belief tree can be combined and the corresponding inequalities can be obtained following the same process. Summing the resulting inequalities will yield (27), leading to (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now focus on the latter of the two terms of (29). We will show the following two properties

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned}
1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
&+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)
\end{aligned} \tag{34}$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \tag{35}$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \tag{36}$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-t) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \tag{37}$$

We apply this as follows to the latter term of (29):

$$\begin{aligned}
&P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\
&\leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\
&\quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\
&= t [P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Eqns 36, 37})
\end{aligned} \tag{38}$$

Applying the same argument to the first term of (29) yields:

$$\begin{aligned}
&P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\
&\leq t [P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1}))]
\end{aligned} \tag{39}$$

Summing (38) and (39) yields (32).

We now complete the proof of (32) and, therefore, the entire proof, by proving (33) and (34).

Proof of (33)

From the recurrence relation in Equation (13) and Equation (3), we can write $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as the product of all belief updates with x_0 , m_0 , and n_0 , respectively, as follows. Each term in the products corresponds to the multiplicative state update after a single action.

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \tag{40}$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \tag{41}$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \tag{42}$$

These equations may be best understood via the illustration in figure 3. We then reorganize terms to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, using the fact that $x_0 = tm_0 + (1-t)n_0$.

$$\begin{aligned}\underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \quad (43)\end{aligned}$$

This proves (33).

Proof of (34)

We first consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We now prove that this can be written as $\alpha_0 m_0 + \beta_0$ for some α_0, β_0 which depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$ and the nature of the dependence is determined by the sequence of actions along this path in stages 1 to $k-1$. We will prove this via induction working backwards from the final term (i.e. the $(k-1)$ -th term) of the product. Therefore our base case will be the final term of this product which can be either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$, depending on the preceding action (in the $(k-1)$ -th stage).

$$\begin{aligned}y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1}))\end{aligned}$$

Thus, the final term of the product is of the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$ where α_{k-1} and β_{k-1} depend only upon $\pi_A^*(x_{k-1})$, and the nature of the dependence is determined entirely by the $(k-1)$ -th action (since this action determines whether the final term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, assume that $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for some $\alpha_{j+1}, \beta_{j+1}$ which depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the nature of the dependence is determined entirely by the sequence of actions in this path for $i > j$ to $i = k-1$. We will now show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_j m_j + \beta_j$ again with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ with the nature of the dependence determined by the sequence of actions on

this path for $i > j-1$ to $i = k-1$. We define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$.

Thus,

$$\begin{aligned}&\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1})\end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, by our inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions in this path from $i > j$ to $i = k-1$. Thus, the claim holds for α_j, β_j , completing the induction.

Now recall that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$. Note that the policy and sequence of actions is the same as those for the numerator. Thus, because, as shown, the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the path of actions, the same logic can be applied to write the denominator

as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 . This can be done similarly for the numerator of the second term in (34). Thus, we can reason about the coefficients in (33) as follows:

$$\begin{aligned}
& t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\
& = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = t m_0 + (1-t) n_0) \\
& = 1
\end{aligned}$$

Thus, (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Assume by way of contradiction that there exists $b \in [0, 1]$ such that $\pi_A^0(b) > \pi_A^*(b)$. We will show that $V_A^*(b) < V_A^\pi(b)$ for a policy π we construct, violating the optimality of $V_A^*(\cdot)$.

Let π be such that, when starting at public belief b , the policy π applies precision $\pi_A^0(b)$ at the current time step and then applies the optimal policy at all future time steps. We now elaborate $V_A^*(b) - V_A^\pi(b)$ by breaking the value function into instantaneous reward r_A and future discounted reward. We will refer to b^+ and b^- as the updated beliefs after receiving signals G and B , respectively when applying the optimal policy and b_m^+ and b_m^- as the same when applying the myopic policy.

$$\begin{aligned}
V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\
&+ \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\
&< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\
&\leq 0 \quad (\text{Theorem 2})
\end{aligned}$$

Here, we applied the fact that the instantaneous reward under $\pi_A^0(\cdot)$ must be greater from the definition of the myopic optimal policy.

We then relied on the convexity of $V_A^*(\cdot)$ from Theorem 2 for the future cost. From Equation (13), note that $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$. Along with this convexity, this leads to the last step above and completes the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ with policy π applying precision q in the current time step and applying the optimal policy at all future time steps. The q that maximizes $V_A^\pi(b)$ provides the optimum precision starting at public belief b . Because the planner never benefits from decreasing precision below the baseline p (see Appendix C.9), we restrict q to $[p, 1]$ without loss of generality.

We will let b^+ and b^- denote the positive and negative belief updates possible after receiving a signal with precision q . Taking the second derivative of $V_A^\pi(b)$ with respect to q for $q \geq \max(b, 1-b)$:

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1-b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1-b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

With the convexity and concavity of $V_A^*(\cdot)$ and $\beta(\cdot)$, respectively, the above shows that $V_A^\pi(b)$ is convex with respect to q on $[\max(b, 1-b), 1]$. Thus, on this interval, it is maximized at one of the two extreme points.

For $q \in [0.5, \max(b, 1 - b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative; thus, the optimal choice will be the baseline precision p to minimize the cost. Thus, there are three possible optimal precisions: p , $\max(b, 1 - b)$, and 1.

Note that the expected value under the latter two candidates is always strictly negative. Under precision p , however, the expected value is 0 at $b = 0$ and $b = 1$. This, combined with the non-positivity and symmetry of the value function (Lemma 8) implies the existence of d_A such that $\pi_A^*(b) = p$ for $b \in [0, d_A] \cup (1 - d_A, 1]$. Furthermore, from Theorem 1 and applying Lemma 9, $d_A \in (0, t_M)$ where t_M is defined as in the statement of Theorem 1.

Finally, Theorem 1 and applying Lemma 9, also imply the existence of $t_A \in [d_A, t_M]$ such that $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$. This completes the proof. ■

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Applying the action rule from (2), the instantaneous reward (6) can be written as follows:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Note that $z(b, q) = b + q - 2bq$ increases with respect to q when $b < 0.5$ and decreases with respect to q when $b > 0.5$.

At $b = 0.5$, $z(b, q) = z(0.5, q) = 0.5$ regardless of the precision chosen. Furthermore, at this public belief, $q \geq \max(b, 1 - b)$ because $q \geq 0.5$ for any q , falling in the first case of (44). In this case, the second term of $r_B(b, q)$ is unchanging with respect to q , so the planner seeks to maximize only the first term, which is accomplished at $q = p$.

We now prove a series of lemmas (10-14) which will assist us in our proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except $q = p$. Thus, we can write its derivative with respect to the chosen precision as follows (except at $q = p$):

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, we can write the second derivative of the reward with respect to q as follows, for $q \neq p$:

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Because $\beta(\cdot)$ is concave, this expression is non-negative, and $r_B(b, q)$ is convex with respect to q . Thus, the reward is maximized at one of the extreme points $\{\max(b, 1 - b), 1\}$ or p because $r_B(b, q)$ is not differentiable at $q = p$. □

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Because $p > 0.5$, for $b \in [0, 1 - p)$, $\max(b, 1 - b) = 1 - b > p$.

First, consider the case when $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. From (44), in such instances, the reward is maximized by minimizing $|q - p|$. Thus, the optimal precision lower than $\max(b, 1 - b)$ is p .

Now consider $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$. What remains is to rule out $\pi_B^0(b) = 1$. To do so, we compare $r_B(b, 1)$ and $r_B(b, 1 - b)$. Both fall under the first case of (44). Because $\beta(\cdot)$ is increasing in its argument and $p < 1 - b \leq 1$, the first term of the reward from (44) is greater for precision $1 - b$ than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$ so we can rule out precision 1. This completes the proof. □

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. The reward is strictly negative for any $q \neq p$ because of the term $-\beta(|q - p|)$. For $b \in (p, 1]$ and $q = p$, $p < \max(b, 1 - b) = b$, leading to the third case of (44) and yielding reward 0. Thus, the optimal precision is p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ contains only $\{0.5\}$. At $b = 0.5$ $p = 1 - b$, thus $\pi_B^0(b) = p = 1 - b = 0.5$. We deal with $p > 0.5 \geq b$ for the remainder of the proof.

For $b \in [1 - p, 0.5]$, $\max(b, 1 - b) = 1 - b \leq p$. Thus, from the first case of (44), precision p yields reward $r_B(b, p) = -Cz(b, p)$. We now consider two cases: $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

Suppose $q < \max(b, 1 - b) = 1 - b$. This leads to the second case of (44) and yields reward $-\beta(|q - p|) - C$. Furthermore, $z(b, q)$ is increasing with respect to q for $b < 0.5$ and takes values between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$. Thus, $z(b, q) < 1$ for all q . Therefore, the reward from policy $q < 1 - b$ is strictly lower than $r_B(b, p)$, and we can rule out this case.

When $q \geq \max(b, 1 - b)$, applying Lemma 10 yields that the remaining candidate precisions are $\{1 - b, p, 1\}$. We rule out precision 1 by comparing it to precision p . The first term of the reward from (44) is greater for precision p than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, the second term is also greater for precision p , and we can rule out precision 1. This completes the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

Note that from the definition, any optimal policy is also ϵ -optimal for any $\epsilon > 0$. Thus, $\pi_B^0(b)$ is also ϵ -optimal

Following this definition, let $\pi_B^{\epsilon,0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or

2. $\pi_B^{\epsilon,0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, $\max(b, 1 - b) = b \leq p$. We consider two cases: $q \geq \max(b, 1 - b)$ or $q < \max(b, 1 - b)$.

In the first case, Lemma 10 yields that $\pi_B^0(b) \in \{b, p, 1\}$. We rule out precision b by comparing it to precision p . The first terms of the reward from (44) is greater for precision p . Furthermore, since $z(b, q)$ is decreasing with respect to q for $b > 0.5$, the second term of the reward is also greater for precision p . Thus, we can rule out precision b . This yields the first claim.

In the latter case, $q < \max(b, 1 - b) = b$. Therefore, (1) since $q \geq 0.5$, $q \in [0.5, b)$, and (2) the third case of (44) applies leading to reward $-\beta(|q - p|)$. Thus, the reward $r_B(b, q)$ is an increasing function of $q \in [0.5, b)$ because $0.5 < b \leq p$. This leads directly to the second claim. $\square$

Remark 3. As noted in Puterman (1990) Section 2.3.1, an optimal policy may not exist for infinite action spaces because the supremum may not be attained. When an optimal policy does not exist, we instead seek ϵ -optimal policies, i.e., policies that yield reward within $\epsilon > 0$ of the supremum. In particular, this is true for case 2 of Lemma 14.

Proof of Theorem 4 We consider the myopic biased policy in 4 cases: $b \in [0, 1 - p]$, $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$, making use of the fact that $p \geq 0.5$. Based on Lemmas 11-14, these cases have the following candidate policies: $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$, respectively.

Case 1: $b \in [0, 1 - p]$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We will show the existence of a threshold t_1 such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

We can express the reward under the two candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

The derivative of $r_B(b, 1 - b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

which is positive as $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Furthermore, when evaluated at $b = 0$ and $b = 1 - p$ we obtain

$$\begin{aligned} r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

Thus, $r_B(b, 1 - b)$ is increasing on $[0, 1 - p]$ with $r_B(b, 1 - b) < r_B(b, p)$ for $b = 0$ and vice versa for $b = 1 - p$. Because $\beta(\cdot)$ is continuous, so are both reward functions. Thus, there exists $t_1 \in (0, 1 - p)$ such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

If $p = 0.5$, this interval contains only $\{0.5\}$, and, for $b = 0.5$, $1 - b = p$. Thus, the two candidate policies are equivalent, and the optimal policy is $\pi_B^0(b) = p$. For the remainder of this case, we consider $p > 0.5$.

We now prove the existence of t_2, t_3 such that $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3]$, and $\pi_B^0(b) = p$ for $b \in [1 - p, t_2] \cup [t_3, 0.5]$.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This leads to the following condition for the optimal policy to be $1 - b$:

$$\begin{aligned} r_B(b, 1 - b) &\geq r_B(b, p) \\ \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1 - p, 0.5]$. The RHS of (46) is a concave parabola that increases in the first half of $[1 - p, 0.5]$ and decreases in the second half.

Now consider the two halves of the interval $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. In the first half, the LHS and RHS of (46) are increasing and concave, with the RHS also being a parabola. Thus, in the first half, the two intersect at most two points (one of which is at $b = 1 - p$). In the second half, the LHS of (46) is increasing while the RHS is decreasing. Thus, they intersect at most once. Let the two possible intersections for $b > 1 - p$ be t_2 and t_3 with $t_2 \leq t_3$. Thus, what remains is to determine the optimal on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1 - b)$ because $p > 0.5$, thus $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. Subsequently, $\pi_B^0(b) = 1 - b$ for (t_2, t_3) , and $\pi_B^0(b) = p$ for $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

In this case, we show the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, and, therefore, $b - \epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now, note that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions in b while $r_B(b, b - \epsilon)$ is convex in b (from the concavity of β). Thus, $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and, as a result, have convex level sets. This implies that the set on which both of these expressions are positive, i.e., the set for which $b - \epsilon$ is the optimal policy, must also be convex. Therefore, $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

What remains is to compare precisions 1 and p on the interval $(0.5, t_5]$. To do so, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 and their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Because $r_B(0.5, 1) < r_B(0.5, p)$ there exists interval $(0.5, t_4)$ with $t_4 \leq t_5$ such that $\pi_B^0(b) = p$. Furthermore, because both are linear in b , there is at most one crossover point where their ordering changes. Thus, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows directly from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward with respect to b can be written as follows:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Because $q \geq 0.5$ and $z(b, q) \in [0, 1]$, it follows that $r_B(b, q)$ is increasing with respect to b .

Next, note that the Bayesian nature of belief updates implies that the belief process is a martingale. That is, for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To demonstrate this consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$, and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the former case, it follows from (3) that $b_{i+1} = b_i$, satisfying the claim.

In the latter case, (13) gives the distribution of b_{i+1} which yields the same.

The monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first prove that the optimal value function is continuous in the following lemma:

Lemma 16. Biased value function continuity

$$V_B^*(\cdot) \text{ is continuous w.r.t. } b.$$

Proof. First consider the one-stage value function defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$. Thus, we can equivalently write the one-stage value function as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1 - b, 1, b - \epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

Using the above expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. First, consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1 - b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of

(44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum, $r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_\epsilon < p$ such that on $[t_\epsilon, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to public belief and the fact that the control variable lies in the compact set $[0.5, 1]$, one might conclude that the optimal is guaranteed to exist at all public beliefs. However, there is an important subtlety which prevents such a conclusion: while $V_B^*(\cdot)$ is continuous with respect to public belief, it need not be continuous with respect to precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)); thus, $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted as $\phi(b_i, q_i)$. Now, b_{i+1} need not be continuous with respect to q_i , thus $\phi(\cdot)$ may not be continuous with respect to q_i . Thus, there may be regimes in which the optimal value is not attained by any policy. Indeed, we will show that such cases exist.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. We again let $z(b, q) = b + q - 2bq$ be the probability of an agent receiving signal B with precision q and prior public belief b and $y(b, q) = 1 - z(b, q)$ be the corresponding probability of the agent receiving signal G . We abuse notation to allow $V_B(b, q)$ to be the expected utility when beginning at public belief b , applying precision q in the current time step, and then following the optimal policy $\pi_B^*(\cdot)$ in all future steps. We can express $V_B(b, q)$ as follows by applying (13), (3), and (44):

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ + \delta y(b, q) V_B^*(f(b, q, G)) \\ + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ -C\mathbf{1}(b < 0.5) \\ + \delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49) when $q < \max(b, 1 - b)$. Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as close to p as possible. Thus, if the optimal precision is less than $\max(b, 1 - b)$, then it will be as close to p as possible.

We now consider four cases: (1) $b \in [0, 1 - p]$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p]$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p]$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p]$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1-b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q-p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1-b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1-\epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q-p|) \rightarrow \beta(1-p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1-p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p-p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1-p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1-p, 0.5)$, $\pi^*(b_i) \geq 1-b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1-b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1-b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1-\delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1-\delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1-b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1-b_i$.

The remaining candidate policies are $q_i \in [1-b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1-b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1-b_i)$ and $b \in [1-p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1-q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1-b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1-b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the

supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b)} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \tag{57}$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q)V_B^*(b, q) \\ &\quad + \delta z(b, q)V_B^*(b, q)] \end{aligned} \tag{58}$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. The one-step welfare term $-C \min(b_i, 1 - b_i, \pi(b_i))$ is monotone increasing in $\pi(b_i)$.

Convexity of social welfare in the public belief follows from the same argument used to prove Theorem 2, because social welfare equals the altruistic value function in the special case $\beta(\cdot) = 0$.

The argument used in Lemma 9 (Appendix C.4) then implies that increasing precision increases welfare. $\square$

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

We show that, for the altruistic planner, the baseline problem is equivalent to a formulation with two observable agent types that have opposing preferences.

Assume two agent types, G and B , and denote agent i 's type by t_i . Type- G agents receive unit reward when $a_i = \omega$ and zero otherwise, while type- B agents receive unit reward when $a_i \neq \omega$ and zero otherwise. Each agent observes her own type and the types of all other agents. All remaining primitives are unchanged.

Agent i 's private belief update is unchanged, because it depends only on the signal structure $\mathbb{P}(s_i = \omega) = q_i$:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_i q_i - b_i - q_i} b_i & s_i = G \\ \frac{1-q}{b_i + q_i - 2b_i q_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), the action rule can be written as follows, where the superscript denotes agent i 's type and $!$ denotes negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because each agent's type is publicly known, $q_i \geq \max(b_i, 1 - b_i)$ implies that observers can infer agent i 's private signal exactly from the action. If $q_i < \max(b_i, 1 - b_i)$, the action is uninformative.

After observing t_i , q_i , and a_i , the public belief therefore updates according to

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this heterogeneous setting, the altruistic planner induces type- G agents to choose ω and type- B agents to choose $!\omega$, because these actions maximize the respective agent utilities.

The planner's instantaneous reward can then be written as follows, with superscript indicating agent i 's type:

$$r^G(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

$$r^B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

The two expressions are identical and coincide with the reward in Equation (5). Thus, with observable heterogeneous types, the altruistic-planner problem is equivalent to the baseline formulation. $\blacksquare$

E EXPERIMENTS

This appendix specifies the experimental implementation used for the LLM-based simulations, including the environment instantiation, parameter grid, prompts, and auxiliary validation checks.

E.1 EXPERIMENTAL SETUP

To simulate controlled social learning, we consider a sequence of agents deciding whether to buy a new car model after observing previous actions and receiving an advertisement whose precision is selected by the planner. LLMs serve three roles: (1) agents, (2) planner, and (3) oracle. Each agent receives a profile containing demographic information, intended car use cases, and preferences. At the beginning of each instance, the offered car is set to either the good or bad type. The information sets and objectives of the planner and agents match the model in Section 3.

The oracle has two functions: (1) generating a generic message that represents the baseline-precision signal in the absence of planner intervention, and (2) generating, after the planner selects a precision, a target-agent-specific message with the prescribed precision.

The oracle receives a fact sheet for each car type, including attributes such as safety rating and customer satisfaction, together with the relevant agent profile. The fact sheet and profile jointly condition generation of a tailored message at the planner-specified precision. Appendix E.2 gives the full experimental and prompting details.

Each agent makes a binary decision from previous decisions and her private signal, with the objective of matching the true state.

The planner observes the same decision history and chooses the precision level for each agent’s message. The planner also receives the precision of a generic untailored message that can be sent at zero cost. Under the altruistic objective, the planner seeks to induce correct decisions; under the biased objective, it seeks to induce car purchases.

We compare LLM-simulation outcomes with analytical solutions computed using JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

The parameterization and prompts below define the exact LLM simulation interface used to instantiate the planner, agents, and oracle in the car-purchase domain.

E.2.1 INSTANCE PARAMETERS

All experiments use the LG AI EXAONE Deep 32B model (LG AI Research, 2025). We solve the analytical model and run the LLM simulations over the parameter grid in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	{0.1, 0.3, 0.5, 0.7, 0.9}
p	Baseline signal precision	{0.6, 0.7, 0.8, 0.9}
δ	Discount Factor	{0, 0.25, 0.5, 0.75, 1}

Table 1: Parameter grid for analytical and LLM experiments

The fact sheets are constructed so that the GOOD car is preferred by all agents and the BAD car is dispreferred by all agents. The corresponding fact sheets are listed below.

```
fact_sheet["GOOD"] = [
    "EPA-rated range: 290 miles.",
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",
    "8-year / 100,000-mile battery warranty.",
    "NHTSA 5-Star overall safety rating.",
    "Cold-soak -5 °F keeps 93 %% range.",
    "Cools cabin 100+72 °F in 3 min.",
    "Cargo volume 26 ft³ total.",
    "Tow's 1 500 lb with factory hitch.",
    "Average 1.2 service visits first 3 yrs.",
    "92 %% owners would buy again (Apr-25).",
    "Qualifies for federal credit.",
    "Quarterly OTA updates incl. battery pre-heat."
]
fact_sheet["BAD"] = [
```

```

1836         "EPA-rated range: 155 mi (combined).",
1837         "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
1838         "Battery warranty: 3 yr / 36 000 mi, first owner only.",
1839         "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
1840         "Cold-soak 15 °F retains 58 %% range; cabin heat cuts range 20%",
1841         "Cabin cools 105 °F → 72 °F in 11 min.",
1842         "Cargo volume 17 ft³; rear seats fixed (no fold-flat).",
1843         "No factory towing approval.",
1844         "Average 3.8 service visits per year (battery-alert issues).",
1845         "Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
1846         "Not eligible for federal tax credit.",
1847         "OTA updates limited to infotainment."
1848     ]

```

Agent profiles were generated programmatically. Each profile includes name, age, city of residence, interests, intended vehicle use case (e.g., commuting or road trips), and major concerns (e.g., fuel efficiency or winter driving).

E.2.2 PROMPTS

The following prompt templates operationalize the information sets, objectives, precision controls, and response formats supplied to each LLM role.

Agent LLM Prompt:

```

1857 sys = (
1858     f"You are deciding whether to buy a car. "
1859     f"Your profile is {profile}"
1860     f"You start believing there's a 50% chance the car is GOOD. "
1861     f"You have two information sources: "
1862     f"1. Previous agents' decisions "
1863     f"2. An advisor's message about the car "
1864     f"Your goal is to maximize your reward "
1865     f"i.e. BUY if the car is GOOD and "
1866     f"NOT BUY if the car is BAD. "
1867     f"If you BUY a GOOD car, you GAIN ${C}. "
1868     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1869     f"If you BUY a BAD car, you LOSE ${C}. "
1870     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1871     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1872     f"decision = exactly 'BUY' or 'NOT BUY'."
1873 )
1874 user = (
1875     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1876     f"ADVISOR MESSAGE:\n{message}\n\n"
1877     f"TASK: Use the previous decisions and the advisor message. "
1878     f"Decide BUY or NOT BUY. "
1879     f"Update your belief about car quality (0.0-1.0). "\n"
1880     f'JSON response:{"belief": <number>,"decision": "<BUY or NOT BUY>"})'
1881 )

```

Planner LLM Prompt:

```

1881 sys = (
1882     f"You are providing messages to potential car buyers. "
1883     f"Each potential car buyer (or agent) has the same information. "
1884     f"Your goal is to get agents to {objective}."
1885     f"Each decision according to your objective, earns you {C} dollars. "
1886     f"You and the agents have the same information. "
1887     f"Initially there is a 50% chance the car is GOOD. "
1888     f"The agents' aim to BUY if the car is GOOD "
1889     f"and NOT BUY if the car is BAD. "
1890     f"You can send a baseline message for free "
1891     f"or tailor the message to the agent at a cost. "
1892     f"Tailoring costs {k} dollars times the change in precision. "

```

```

1890     f"(Message precision is the probability (0.5 to 1.0) "
1891     f"that the message leaves the buyer with the correct impression. "
1892     f"For example, if the car is likely GOOD: "
1893     f"a high precision message will likely suggest BUY "
1894     f"and a low precision message will likely suggest NOT BUY. "
1895     f"You may choose to increase or decrease precision. "
1896     f"Agents decide based on your message and the decision history. "
1897     f"They update beliefs by observing previous decisions. "
1898     f"IMPORTANT: Thus your objective is to maximize:"
1899     f"{C} * (# of agents who choose according to your objective) "
1900     f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1901     f"where {d} is your discount factor for future cost. "
1902 )
1903 user = (
1904     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1905     f"DECISION HISTORY:\n{history_txt}\n"
1906     f"TASK:"
1907     f"Based on the history and your objective, "
1908     f"choose your desired precision.\n"
1909     f'JSON response: {"message_precision": <number>}\n'
1910 )

```

Oracle LLM Prompt:

```

1910 sys = (
1911     f"You will be given "
1912     f"1. an agent profile, "
1913     f"2. information about a car, "
1914     f"3. the true quality of the car, "
1915     f"4. a baseline message and the 'precision' of that message, and "
1916     f"5. a desired precision for the message you output. "
1917     f"(Message precision is the probability (0.5 to 1.0) "
1918     f"that the message leaves the buyer with the correct impression. "
1919     f"For example, if the car is likely GOOD: "
1920     f"a high precision message will likely suggest BUY "
1921     f"and a low precision message will likely suggest NOT BUY. "
1922     f"IMPORTANT: Output a message with the requested precision. "
1923     f"IMPORTANT: The message should be tailored to the profile. "
1924 )
1925 user = (
1926     f"AGENT PROFILE:{profile}\n"
1927     f"TRUE CAR QUALITY: {omega}\n\n"
1928     f"CAR FACTS: {car_facts}\n"
1929     f"BASELINE MESSAGE:{baseline_message}\n"
1930     f"BASELINE MESSAGE precision: {p}\n"
1931     f"DESIRED MESSAGE precision: {q}\n"
1932     f"IMPORTANT: Output a message with the requested precision. "
1933     f"TASK: Write a message (MAX 100 words) with the desired precision."
1934     f'JSON response: {"message": "<message>"}\n'
1935 )

```

E.3 VALIDATION OF LLM BELIEFS AND ORACLE

We report additional experiments validating the LLM self-reported beliefs used in figure 1b and the LLM oracle's precision control. These experiments treat LLMs as rational utility-maximizing agents that need not update according to Bayes' rule. The objectives are (1) to test whether the self-reported posterior corresponds to the belief used in state-contingent choices, and (2) to verify that oracle-generated messages realize the requested precision.

First, we provide the LLM with a prior and a new signal, elicit its self-reported posterior, and then ask for a binary "buy/not-buy" decision that yields unit reward if and only if the true state is good. A rational agent buys exactly when its subjective belief exceeds 0.5. As shown in figure 4a, the LLM's decisions closely follow this threshold: it usually buys above 0.5 and does not buy below 0.5. Second, we elicit willingness to pay for a state-contingent payoff. For prices c , we ask whether the agent would pay c in exchange for unit utility if the state is good. A rational agent with subjective belief p accepts exactly when $c < p$. Figure 4b reports the

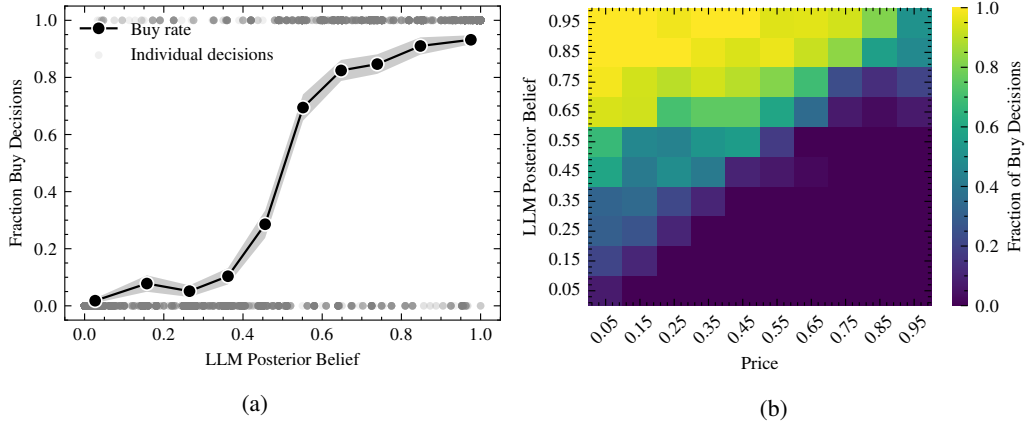


Figure 4: LLM buy-decision rate as a function of (a) self-reported posterior belief and (b) self-reported posterior together with purchase price.

fraction of “buy” decisions across posterior–price pairs and shows the expected separation: acceptance is high for $c < p$ and low for $c > p$. Together, the two experiments indicate that the self-reported posterior closely tracks the belief used by the model in decision making.

We also validate oracle precision control. After the oracle generates a message at specified precision q , a separate LLM estimates the realized precision $\hat{q}$. We evaluate $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ across 100 programmatically generated profiles with 10 trials per combination, for 6,000 total trials. The mean absolute error $|q - \hat{q}|$ is 0.025, indicating that oracle messages closely track the requested precision.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We examine belief trajectories from a representative LLM experiment to illustrate the aggregate effect of non-Bayesian updating. The policy is fixed to the analytically optimal policy and then applied to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0, whereas the LLM belief (solid black line) changes more slowly. This aggregate pattern follows from the individual-level biases: public belief resists entering a cascade (NB3) because agents overreact to occasional positive signals that contradict the increasingly negative group belief (NB2).

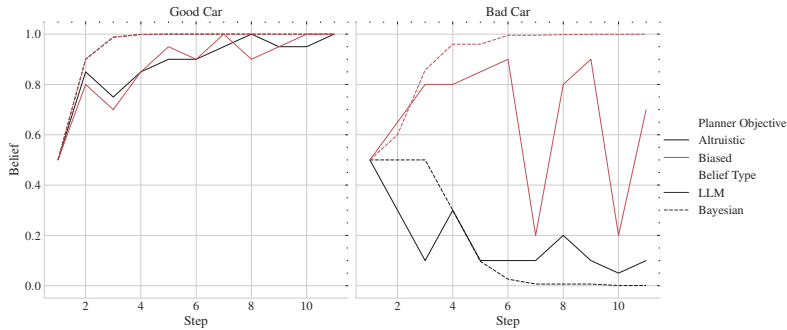


Figure 5: Belief-update trajectories for the Bayesian public belief and the LLM-reported belief under analytically optimal policies, shown across planner objectives and true states.